



CS 3151 - Software Engineering

Final Project Report

By: Dana Mounayer & Jwdee Alamam

Instructor: Dr. Passent Elkafrawy

Spring 2026

Abstract.....	4
1. Introduction.....	4
1.2 Purpose.....	4
1.3 Definitions and Acronyms.....	4
1.4 Scope.....	5
1.5 Comparative Table of Reviewed Systems.....	5
1.6 Technologies Used in Wearable Monitoring Systems.....	6
1.7 Problem Statement and Research Gaps.....	6
2. Project Timeline.....	6
3. System Requirements.....	7
3.1 Functional Requirements.....	7
3.2 Non-Functional Requirements.....	8
4. Use Case Analysis.....	8
4.1 Actors.....	8
4.2 User Stories.....	8
4.3 Use Case List.....	9
4.4 Traceability Matrix.....	9
4.5 Detailed Use Cases.....	10
UC1 – Monitor Patient Status.....	10
UC2 – Manage Seizure Records.....	11
UC3 – Emergency Alert & Response.....	12
UC4 – Clinical Review & Report Management.....	14
5. System Architecture.....	15
5.1 System Architecture Overview & Diagram.....	15
5.2 Context Diagram.....	16
5.4 System Data Flow Diagram.....	16
5.5 Architectural Design Decisions.....	17
5.6 Scalability and Reliability Considerations.....	18
6. System Design.....	18
6.1 Design Approach.....	18
6.2 Key Design Components.....	18
6.3 Design Patterns.....	18
7. User Interface Design.....	19
7.1 Main Screens.....	19
7.2 Usability Considerations.....	20
7.3 External Interfaces.....	20
8. Class Diagram.....	20
9. ER Diagram.....	21

10. Risk Management.....	22
11. Conclusion.....	22
11.1 Future Work.....	22
12. Code Script.....	22
13. References.....	25

Abstract

Epilepsy is a neurological disorder characterized by unpredictable seizures that affect patient safety and quality of life. Traditional seizure monitoring methods rely mainly on hospital-based electroencephalography (EEG), which is expensive, time-limited, and unsuitable for continuous monitoring in daily life. This project proposes a Smart Wearable Seizure Monitoring System that integrates multi-sensor physiological data, including EEG, heart rate, stress, and motion, with artificial intelligence to support real-time seizure detection and automated caregiver notification. The system applies modern software engineering practices such as Agile development, DevOps pipelines, secure system design, and AI-driven software engineering to ensure reliability, scalability, maintainability, and security. By combining wearable technologies, distributed architectures, intelligent data processing, and structured reporting, the proposed solution enables continuous monitoring, timely intervention, and clinically useful seizure analytics for patients, caregivers, and neurologists.

1. Introduction

Epilepsy is a neurological condition marked by unpredictable seizures that may lead to injury and delayed emergency response. Continuous monitoring is therefore essential for improving diagnosis, treatment planning, and patient safety. However, traditional seizure monitoring approaches based on hospital EEG recordings are limited in duration, costly, and impractical for long-term real-world use.

This project introduces a Smart Wearable Seizure Monitoring System that combines wearable sensing technologies with a mobile application and cloud-based processing to monitor patient data continuously, detect abnormal seizure-related patterns, notify caregivers immediately, and provide neurologists with organized clinical reports. The goal is to improve patient safety, reduce response time during seizure events, and support more informed clinical decision-making.

1.2 Purpose

The purpose of this system is to develop a smart wearable seizure monitoring solution that detects seizures in real time and notifies caregivers immediately. The system is designed to support patients, caregivers, and neurologists by improving safety, monitoring, and clinical reporting.

1.3 Definitions and Acronyms

The following definitions and acronyms are used throughout this report:

- EEG (Electroencephalography): A method used to record electrical activity of the brain.
- FR (Functional Requirement): A requirement describing what the system should do.
- NFR (Non-Functional Requirement): A requirement describing system qualities such as performance and security.
- TLS (Transport Layer Security): A protocol used to secure data during transmission.

1.4 Scope

The proposed system includes the following:

- Collection and continuous monitoring of physiological signals using wearable sensors
- AI-based seizure detection and abnormal pattern analysis
- Real-time alert notification to registered caregivers
- Logging and visualizations of seizure events
- Generation of structured reports for neurologists

1.5 Comparative Table of Reviewed Systems

Study / System	Dataset Size	Method / SE Approach	Accuracy	Deployment Type
EPILAB (Teixeira et al., 2011)	Variable research datasets with long-term EEG/ECG	Research-focused MATLAB toolbox, experimental and exploratory	N/A	Research environment
Wearable Sensor Monitoring (Patel et al., 2012)	Multiple wearable monitoring studies	Review of wearable sensor architectures for physiological monitoring	N/A	Wearable healthcare monitoring
Persyst / Encevis / BESA (Reus et al., 2025)	Large dataset, 286 EEG recordings and 8,771 hours	Clinical EMU workflow, structured review	Approximately 84–98%	EMU, live clinical observation
Encevis 2.0 / Persyst 14 (Koren et al., 2025)	Medium dataset, 80 patients and 6,826 hours	FDA/CE-certified clinical software with iterative updates	Approximately 68–89%	EMU / clinical use

Darankoum et al. (2026)	Small to medium dataset, animal data with human EEG validation	Deep learning pipeline using CNN, RNN, and attention models	Approximately 93%	Research / experimental
--------------------------------	--	---	-------------------	-------------------------

1.6 Technologies Used in Wearable Monitoring Systems

The proposed Smart Wearable Seizure Monitoring System uses several technologies to support continuous monitoring, seizure detection, communication, and reporting. Artificial intelligence is used to analyze physiological signals and detect seizure-related patterns in real time. Wearable sensors collect data such as EEG, heart rate, stress, and motion, while a distributed system connects wearable devices, mobile applications, cloud servers, and notification services to support real-time communication and remote monitoring. DevOps and containerization can also support reliable deployment, maintenance, and scalability.

1.7 Problem Statement and Research Gaps

Many existing systems focus on hospital-based EEG infrastructure or invasive medical devices. As a result, they are often not suitable for continuous real-world monitoring, accessible daily use, or direct caregiver involvement. In addition, many systems focus heavily on detection algorithms while providing limited support for mobile-centered emergency workflows, personalized monitoring, secure distributed communication, and structured analytics for clinicians.

These gaps motivate the development of a smart wearable solution that combines AI, wearable sensing, distributed architecture, and secure software engineering practices into one integrated monitoring platform.

2. Project Timeline

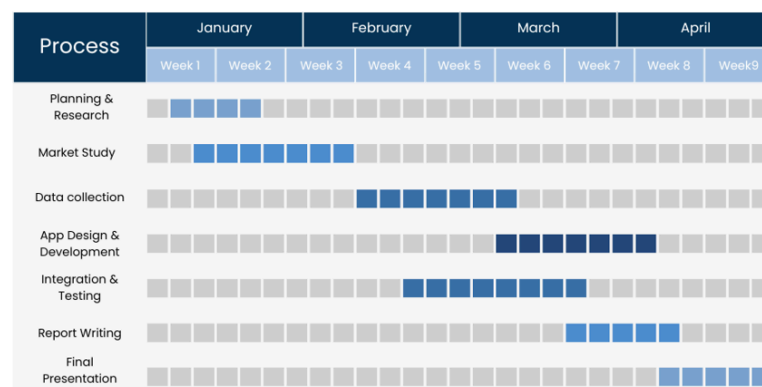


Figure 1: Project Timeline for the Smart Wearable Seizure Monitoring System

3. System Requirements

3.1 Functional Requirements

ID	Requirement Name	Description
FR1	Seizure Detection	The system shall analyze EEG, heart rate, stress, and motion data at a minimum sampling rate of 1 Hz and detect seizure activity with a target accuracy of up to 99%.
FR2	Event Logging	The system shall record seizure start time and end time with ± 1 second precision and store them in the patient database immediately after detection.
FR3	Frequency Calculation	The system shall automatically update seizure frequency statistics (daily, weekly, monthly) within 3 seconds after each detected event.
FR4	Alert Delivery	The system shall send push notifications to at least two registered caregivers and trigger SMS fallback if delivery fails within 3 seconds, with total alert delivery time not exceeding 5 seconds.
FR5	Emergency Call Feature	The system shall allow caregivers to initiate an emergency call directly from the application with one tap.
FR6	Data Filtering	The system shall allow patients and neurologists to filter seizure history using custom date ranges.
FR7	Report Export	The system shall generate downloadable PDF medical reports including seizure duration, frequency statistics, and timestamps within 5 seconds of request.

3.2 Non-Functional Requirements

ID	Requirement Name	Description
NFR1	Reliability	The system shall maintain at least 99% uptime per month, excluding scheduled maintenance not exceeding 2 hours.
NFR2	Security	All patient data shall be encrypted using AES-256 encryption at rest and TLS 1.3 during transmission.
NFR3	Availability	The system shall operate continuously (24/7 monitoring) while the wearable device is active and connected.
NFR4	Usability	The application shall allow users to access seizure history within a maximum of 3 navigation steps.
NFR5	Performance	The system shall process incoming physiological data and update outputs within 2 seconds.
NFR6	Scalability	The system shall support multiple users concurrently without degradation in performance.

4. Use Case Analysis

4.1 Actors

The system involves multiple types of users interacting with different system components:

- **Patient:** Monitors health status and seizure history.
- **Caregiver:** Receives alerts and responds to emergencies.
- **Neurologist:** Reviews patient data and reports.
- **System:** Processes data, detects seizures, and triggers alerts.

4.2 User Stories

- As a **patient**, I want automatic seizure detection so that I remain safe even when I am alone.
- As a **caregiver**, I want to receive instant alerts so that I can respond quickly to emergencies.
- As a **neurologist**, I want structured reports so that I can make informed clinical decisions.

- As a **user**, I want a simple and clear interface so that I can easily access important information.

4.3 Use Case List

ID	Use Case	Primary Actor	Goal	Priority	Related Requirements
UC1	Monitor Patient Status	Patient / Caregiver	View real-time physiological data	Must	FR1, FR3
UC2	Manage Seizure Records	Patient	Record and review seizure data	Must	FR2, FR3, FR6
UC3	Emergency Alert & Response	System / Caregiver	Notify caregivers and enable response	Must	FR4, FR5
UC4	Clinical Review & Reports	Neurologist	Analyze and generate reports	Must	FR3, FR7

4.4 Traceability Matrix

<i>Requirement</i>	<i>UC1</i>	<i>UC2</i>	<i>UC3</i>	<i>UC4</i>
<i>FR1</i>	✓		✓	
<i>FR2</i>		✓		✓
<i>FR3</i>	✓	✓		✓
<i>FR4</i>			✓	
<i>FR5</i>			✓	
<i>FR6</i>		✓		
<i>FR7</i>				✓

4.5 Detailed Use Cases

UC1 – Monitor Patient Status

Primary Actors: Patient / Caregiver

Preconditions: Wearable device connected; user authenticated

Postconditions: Real-time physiological data displayed

Main Flow:

1. The user opens the mobile application.
2. The system displays real-time EEG, heart rate, stress, and motion data.
3. The system highlights abnormal patterns if detected.

Alternate Flow:

- If the device is disconnected, the system displays a connection error message.

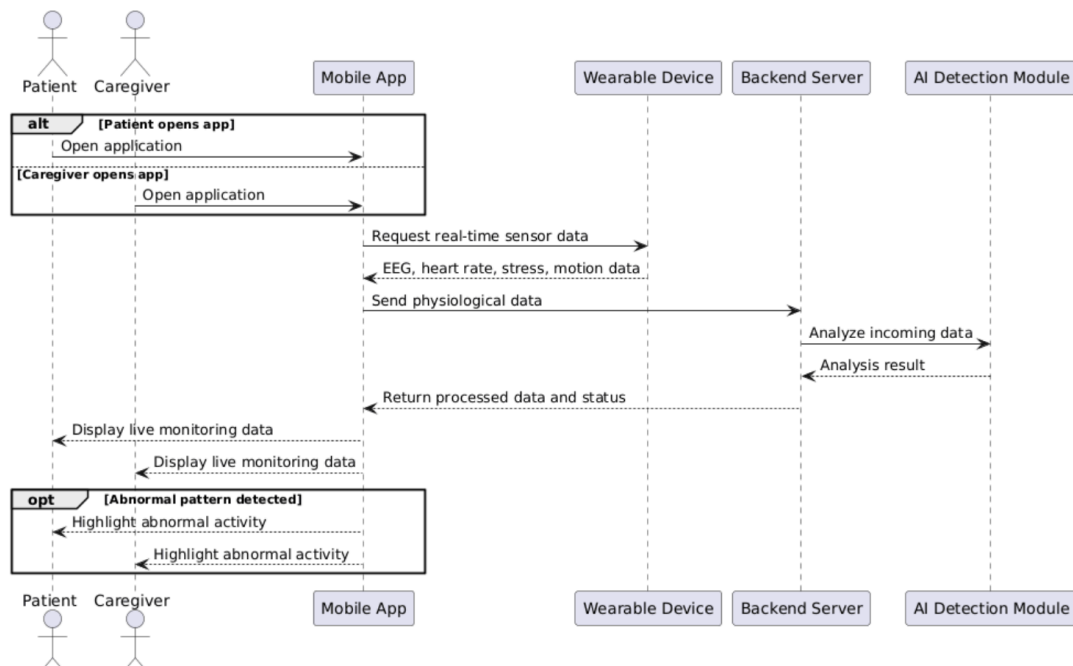


Figure 2: Sequence Diagram for UC1 – Monitor Patient Status.

Process Description

1. The patient or caregiver opens the mobile application.
2. The application requests real-time physiological data from the wearable device.
3. The wearable device sends EEG, heart rate, stress, and motion data.
4. The application forwards the data to the backend server.
5. The backend server sends the data to the AI detection module for analysis.
6. The AI module returns the monitoring result.
7. The application displays the processed data and current status to the user.

UC2 – Manage Seizure Records

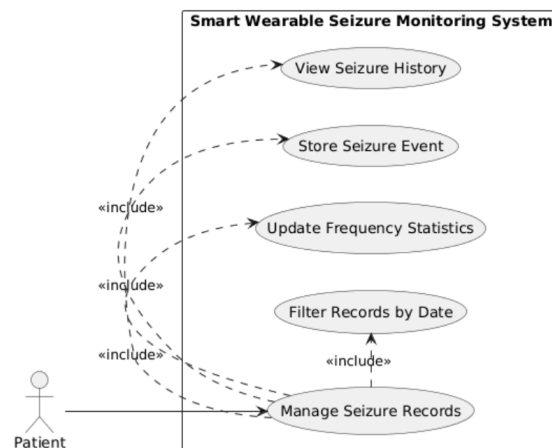
Primary Actor: Patient

Preconditions: At least one seizure event exists

Postconditions: Seizure data stored and updated

Main Flow:

1. The patient accesses seizure history.
2. The system displays recorded seizure events.
3. The system updates frequency statistics.
4. The user filters data by date if needed.



Alternate Flow:

- If no records exist, the system displays an empty state message.

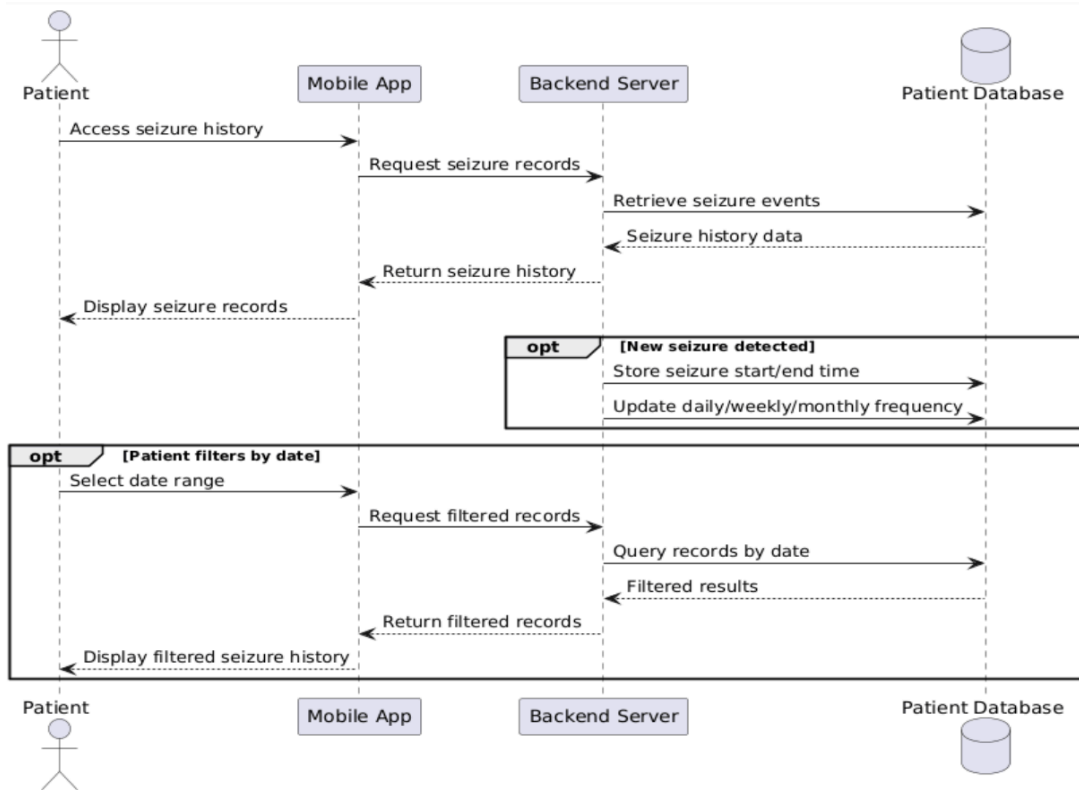


Figure 3: Sequence Diagram for UC2 – Manage Seizure Records.

Process Description

1. The patient opens the seizure history section in the mobile application.
2. The application requests stored seizure records from the backend server.
3. The backend server retrieves the seizure history from the database.
4. The application displays the recorded seizure events to the patient.
5. Newly detected events are stored, and seizure frequency statistics are updated.
6. If needed, the patient can filter records by date range.

UC3 – Emergency Alert & Response

Primary Actor: System

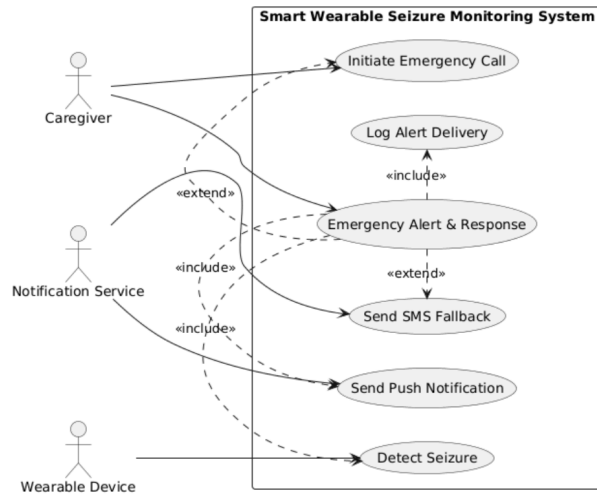
Secondary Actor: Caregiver

Preconditions: Seizure detected; device connected

Postconditions: Caregivers notified; alert logged

Main Flow:

1. The system detects seizure activity using AI analysis.
2. The system prepares an alert notification.
3. Push notifications are sent to caregivers.
4. SMS fallback is triggered if needed.
5. The caregiver receives the alert.
6. The caregiver initiates an emergency call if required.
7. The system logs the event.



Alternate Flow:

- If notification fails, the system retries and logs failure status.

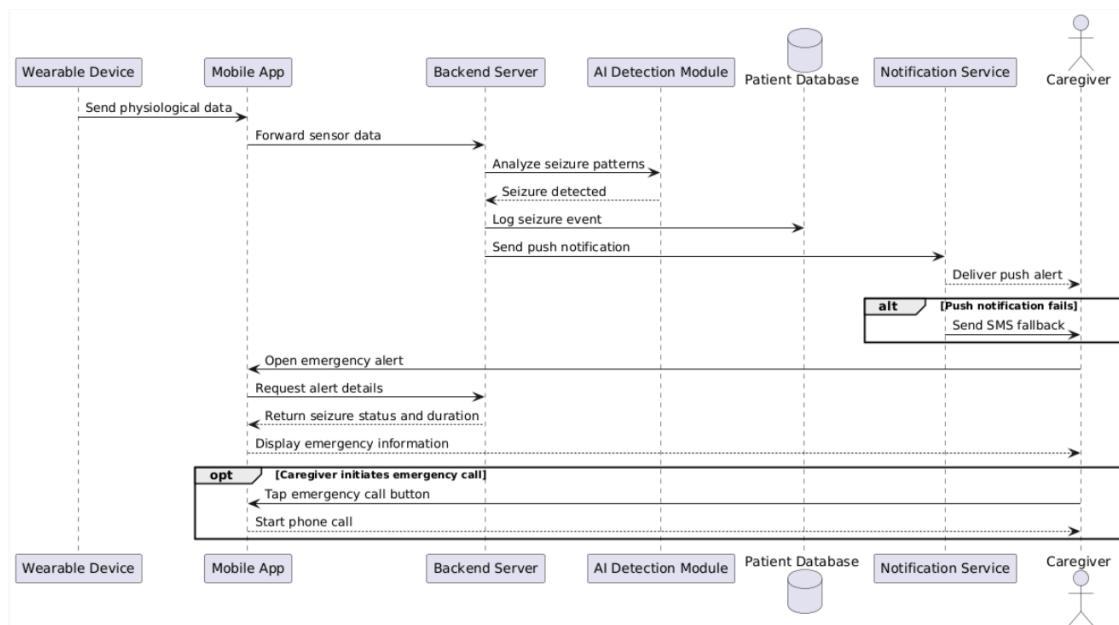


Figure 4: Sequence Diagram for UC3 – Emergency Alert & Response.

Process Description

1. The wearable device collects physiological data from the patient.
2. The data is sent to the mobile application and forwarded to the backend server.
3. The backend server sends the data to the AI detection module for seizure analysis.
4. If seizure activity is detected, the system records the event in the database.
5. The system sends an alert to the caregiver through the notification service.

6. SMS fallback is triggered if needed.
7. The caregiver views the alert and can initiate an emergency call.

UC4 – Clinical Review & Report Management

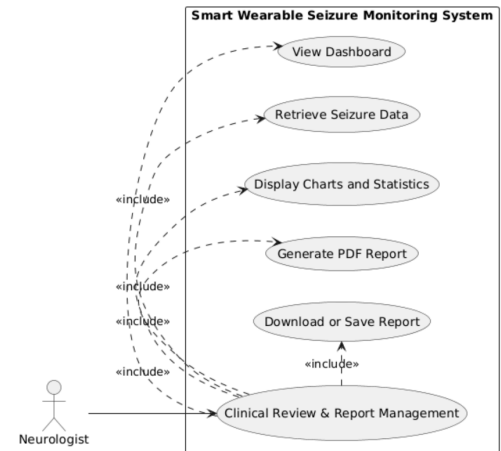
Primary Actor: Neurologist

Preconditions: Seizure data available; user authenticated

Postconditions: Report generated and accessible

Main Flow:

1. The neurologist accesses the dashboard.
2. The system retrieves patient data.
3. The system calculates statistics and displays charts.
4. The neurologist generates a report.
5. The report is downloaded or saved.



Alternate Flow:

- If data is unavailable, the system displays an error message.

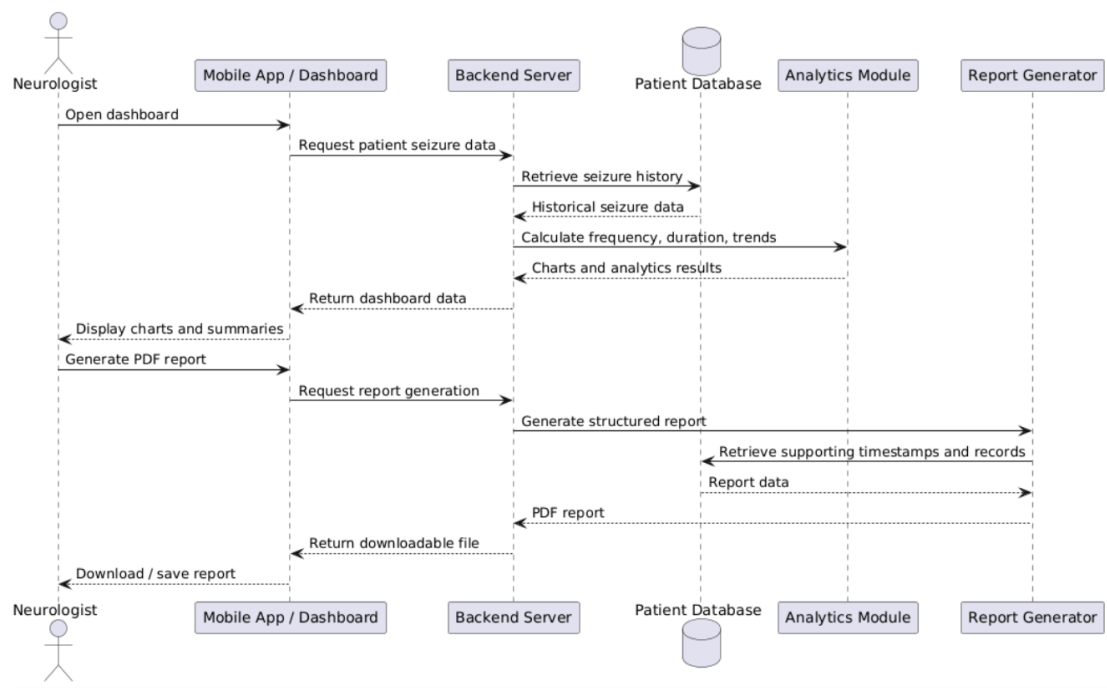


Figure 5: Sequence Diagram for UC4 – Clinical Review & Report Management.

Process Description

1. The neurologist opens the dashboard in the application.
2. The application requests patient seizure data from the backend server.
3. The backend retrieves the records from the database and sends them to the analytics module.
4. The system generates seizure summaries, charts, and trends.
5. The application displays the analytics to the neurologist.
6. The system generates a PDF report when requested and sends it for download.

5. System Architecture

5.1 System Architecture Overview & Diagram

The Smart Wearable Seizure Monitoring System follows a layered distributed architecture. The presentation layer consists of the mobile application, which provides the interface for patients, caregivers, and neurologists. The application layer is represented by the backend server, which coordinates requests, manages business logic, and controls communication between components. The data processing layer includes the AI detection module, analytics module, and report generator, which analyze physiological signals, detect seizure activity, calculate statistics, and produce structured medical reports. The data layer stores patient information and seizure records in the database. The integration layer connects the system to the wearable device for real-time physiological data collection and to the notification service for caregiver alerts. This architecture improves modularity, maintainability, and scalability by assigning each layer a clear responsibility.

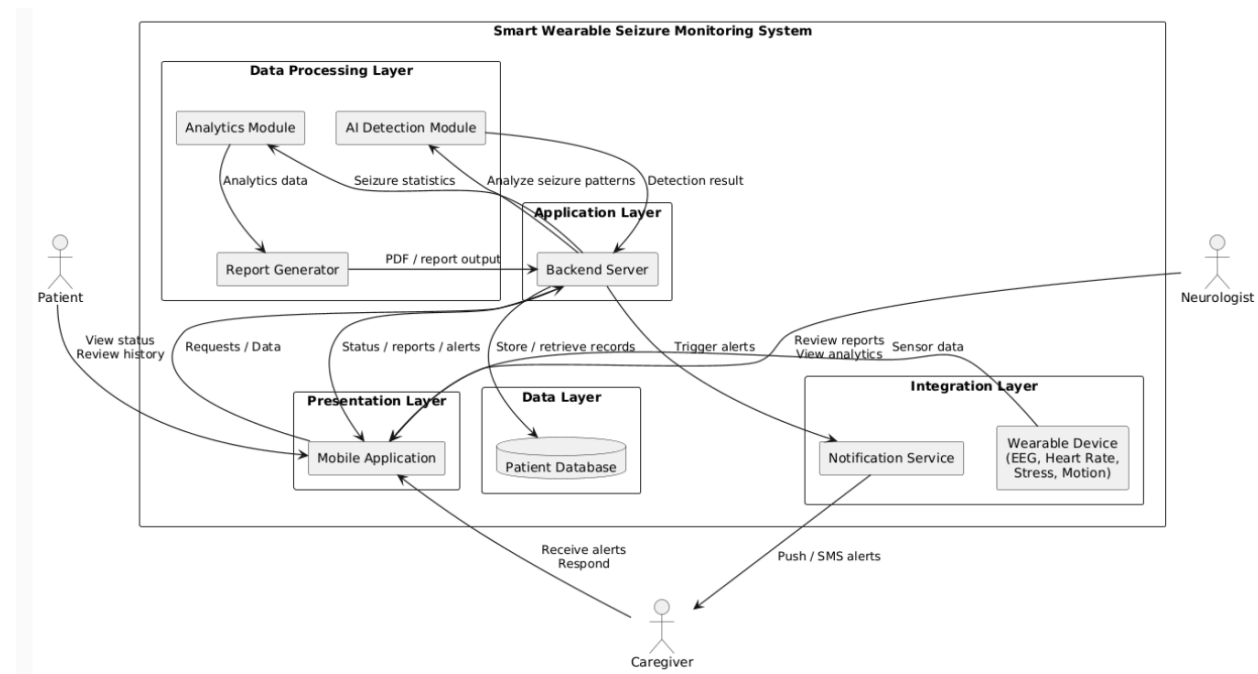


Figure 6: System Architecture Diagram for the Smart Wearable Seizure Monitoring System

5.2 Context Diagram

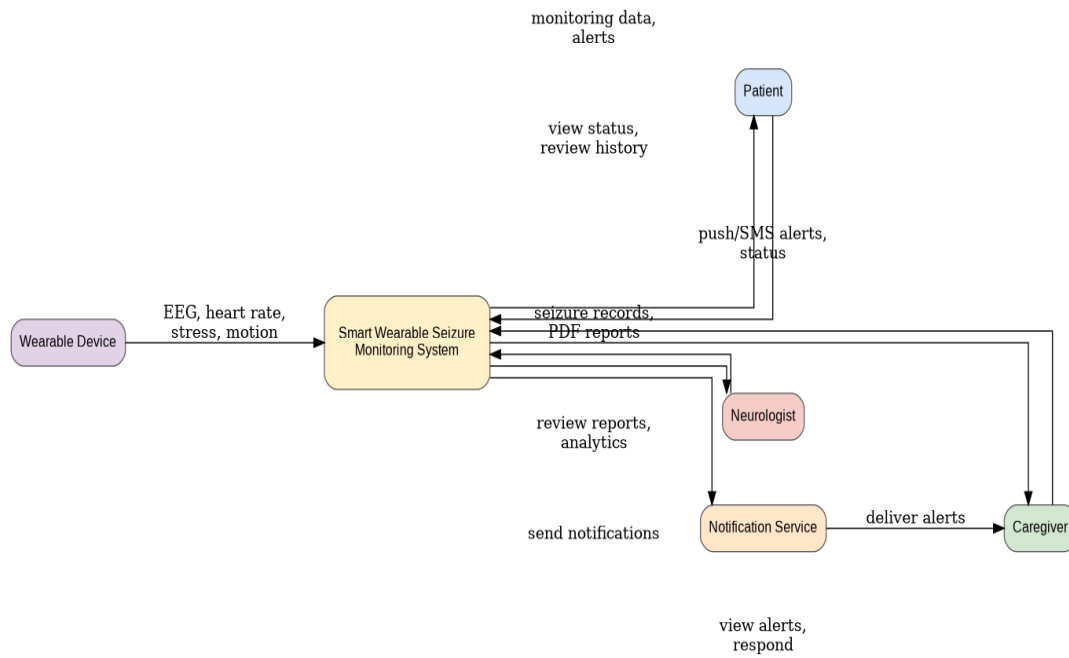


Figure 7: Context Diagram for the Smart Wearable Seizure Monitoring System

5.4 System Data Flow Diagram

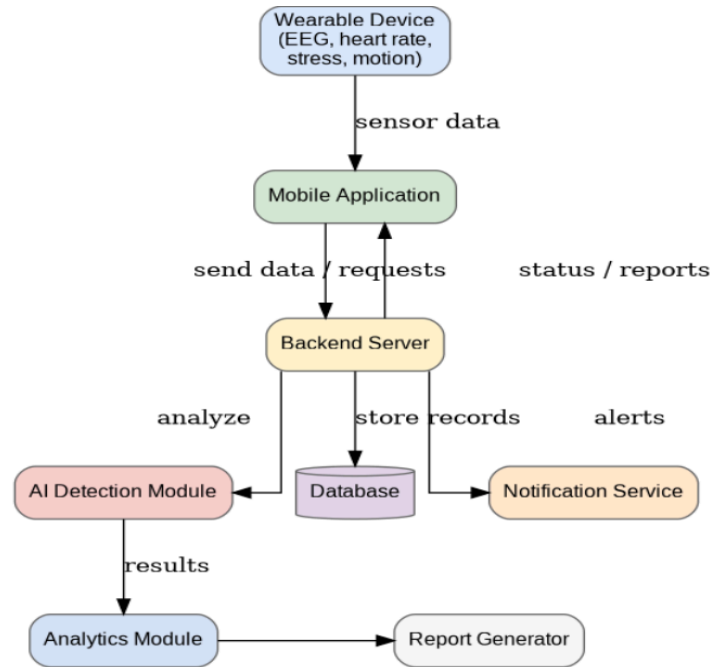


Figure 8: Data Flow Diagram for the Smart Wearable Seizure Monitoring System

Data Flow Description

The wearable device continuously collects physiological data from the patient.

The data is transmitted to the mobile application in real time.

The mobile application forwards the data to the backend server.

The backend server sends the data to the AI detection module for analysis.

If a seizure is detected, the system logs the event in the database.

The backend server triggers the notification service to alert caregivers.

The system updates seizure statistics using the analytics module.

The report generator creates structured reports based on stored data.

Users access results and reports through the mobile application.

5.5 Architectural Design Decisions

The following design decisions were made to ensure system efficiency and reliability:

- **Distributed Architecture**
Enables real-time monitoring and remote access across multiple devices.
- **AI-Based Processing**
Improves detection accuracy and allows intelligent analysis of physiological data.

- **Cloud-Based Backend**
Supports scalability, data storage, and continuous availability.
- **Modular Design**
Allows independent development and maintenance of system components.
- **Secure Communication**
Ensures patient data protection using encryption and secure protocols.

5.6 Scalability and Reliability Considerations

The system is designed to handle increasing numbers of users and continuous data streams by:

- Using cloud infrastructure to scale backend services dynamically
- Separating processing modules to reduce system overload
- Implementing fallback mechanisms such as SMS alerts
- Ensuring high availability through reliable server deployment

6. System Design

The system design defines how the components of the Smart Wearable Seizure Monitoring System interact to support monitoring, detection, alerting, and reporting. The design follows a modular structure that separates data collection, processing, and user interaction.

6.1 Design Approach

The system follows a layered and modular design approach. Each component is responsible for a specific function, which improves system organization, scalability, and maintainability.

6.2 Key Design Components

- **Wearable Device:** Collects physiological data from the patient.
- **Mobile Application:** Displays data, alerts, and reports to users.
- **Backend Server:** Processes data and coordinates system operations.
- **AI Detection Module:** Analyzes data and detects seizure activity.
- **Database:** Stores patient data and seizure records.
- **Notification Service:** Sends alerts to caregivers.

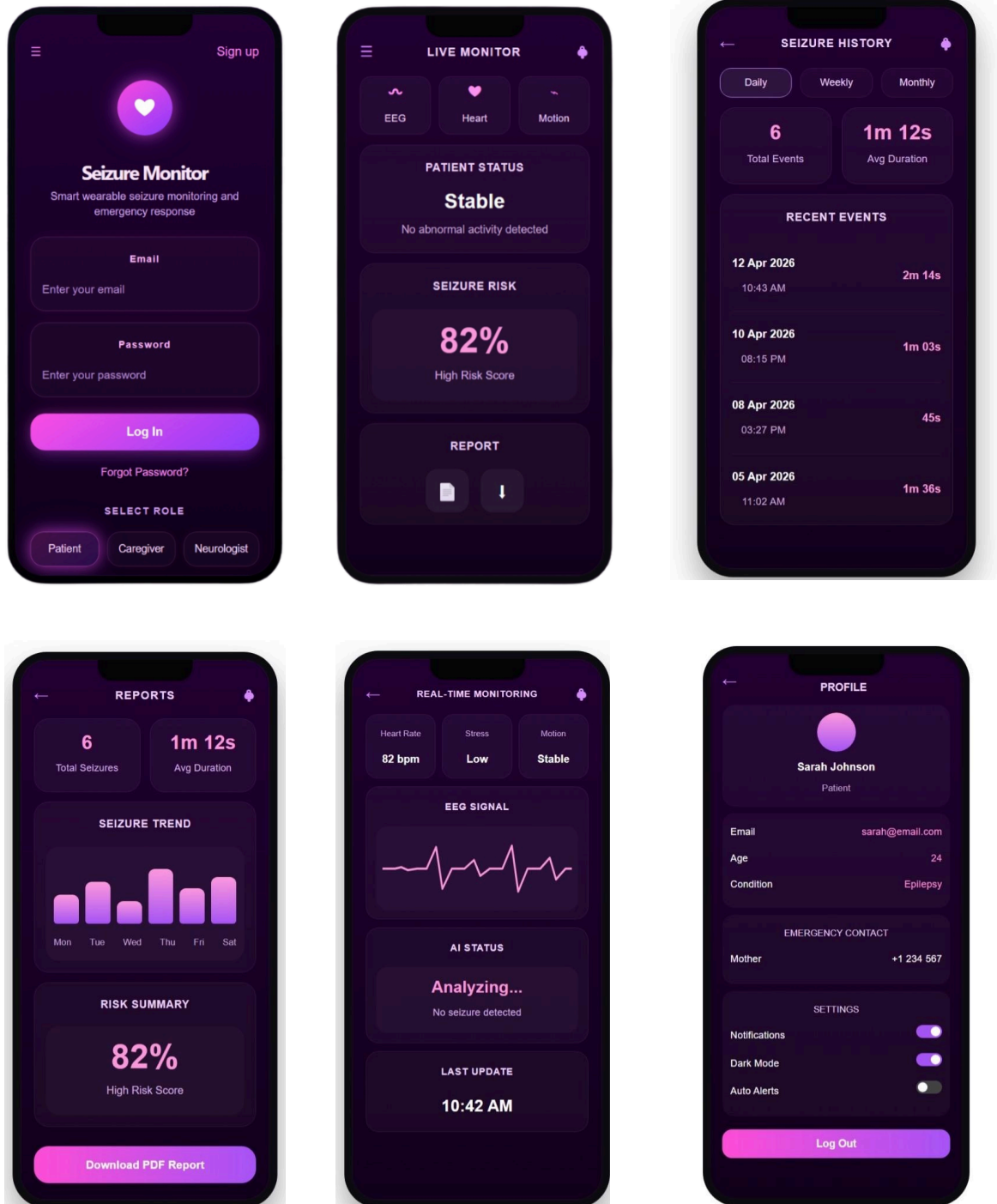
6.3 Design Patterns

- **Observer Pattern:** Used for real-time notifications and alerts.
- **MVC Pattern:** Used to separate user interface, logic, and data handling.

7. User Interface Design

The user interface is designed to be simple, clear, and easy to use for all users, especially patients and caregivers during emergency situations. The design focuses on readability, quick access to important features, and minimal interaction steps.

7.1 Main Screens



7.2 Usability Considerations

The interface is designed for users with different technical levels. Important features are easy to access, and emergency actions can be performed quickly. The design also ensures clear visibility of alerts and system status.

7.3 External Interfaces

The system interacts with multiple external components as described below:

- **User Interfaces:**
The system provides a mobile application interface that allows patients, caregivers, and neurologists to interact with the system, view data, receive alerts, and generate reports.
- **Hardware Interfaces:**
The system connects with wearable devices equipped with sensors that collect physiological data such as EEG, heart rate, and motion.
- **Software Interfaces:**
The system communicates with backend servers, databases, and AI modules through APIs to process data and manage system operations.
- **Communication Interfaces:**
The system uses standard communication protocols such as HTTP/HTTPS and TCP/IP to ensure secure and reliable data transmission. Notification services are used for push notifications and SMS alerts.

8. Class Diagram

The class diagram represents the static structure of the Smart Wearable Seizure Monitoring System. It identifies the main classes of the system, their attributes, methods, and relationships.

The main classes in the system include Patient, Caregiver, Neurologist, Sensor Data, Seizure Event, Alert, Report, and AI Model. These classes reflect the essential entities and processes involved in continuous monitoring, seizure detection, emergency alerting, and clinical reporting.

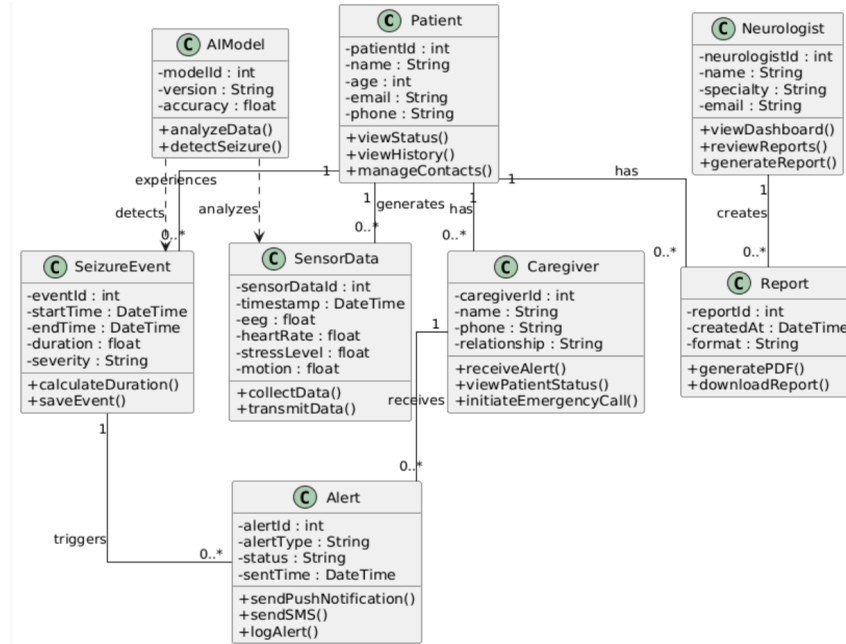


Figure 8: Class Diagram for the Smart Wearable Seizure Monitoring System

9. ER Diagram

The Entity-Relationship (ER) diagram represents the structure of the system database. The system includes key entities such as Patient, Caregiver, Seizure Event, Sensor Data, Alert, Report, and Neurologist.

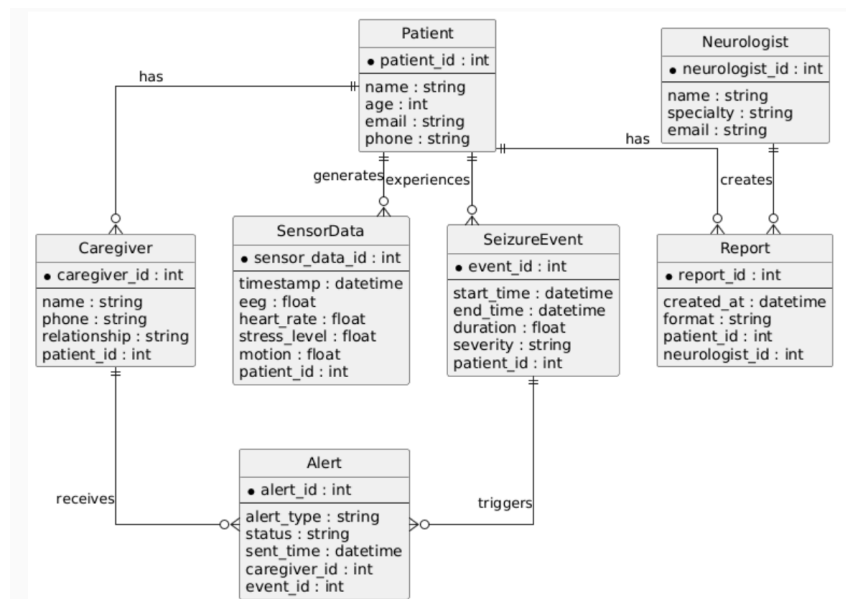


Figure 9: Entity-Relationship Diagram for the Smart Wearable Seizure Monitoring System

10. Risk Management

Risk	Impact	Mitigation
Sensor failure	High	Use reliable hardware and fallback mechanisms
Network failure	High	Implement offline buffering and SMS fallback
Data privacy issues	High	Apply encryption and secure protocols
AI model inaccuracy	Medium	Continuous training and validation
System overload	Medium	Use scalable cloud infrastructure

11. Conclusion

The Smart Wearable Seizure Monitoring System provides an intelligent solution for monitoring patients and improving response time during seizures. By integrating wearable technology, mobile applications, and AI-based detection, the system enhances patient safety and supports caregivers and medical professionals.

The system offers real-time monitoring, automatic alerts, and detailed reports, making it a valuable tool in healthcare.

11.1 Future Work

Future improvements of the system may include:

- Integrating with smartwatches and additional wearable devices
- Adding GPS tracking for precise location detection
- Supporting more health monitoring features

12. Code Script

```
import './App.css';

function App() {
  return (
```

```

<div className="page">
  <div className="phone-frame">
    <div className="notch"></div>

    <div className="screen profile-screen">
      <div className="top-bar">
        <span className="menu-icon"></span>
        <span className="top-title">PROFILE</span>
        <span className="bell-icon">
          <span className="bell-body"></span>
          <span className="bell-top"></span>
          <span className="bell-dot"></span>
        </span>
      </div>

      { /* Profile */ }
      <div className="profile-card">
        <div className="avatar"></div>
        <div className="user-name">Sarah Johnson</div>
        <div className="user-role">Patient</div>
      </div>

      { /* Info */ }
      <div className="info-card">
        <div className="info-row">
          <span>Email</span>
          <span className="info-value">sarah@email.com</span>
        </div>
        <div className="info-row">
          <span>Age</span>
          <span className="info-value">24</span>
        </div>
        <div className="info-row">
          <span>Condition</span>
          <span className="info-value">Epilepsy</span>
        </div>
      </div>

      { /* Emergency */ }
      <div className="settings-card">
        <div className="card-title">EMERGENCY CONTACT</div>
        <div className="setting-row">
          <span>Mother</span>
          <span className="setting-value">+1 234 567</span>
        </div>
      </div>
    </div>
  </div>
</div>

```

```

    { /* Settings */}
    <div className="settings-card">
      <div className="card-title">SETTINGS</div>

      <div className="setting-row">
        <span>Notifications</span>
        <div className="toggle on"></div>
      </div>

      <div className="setting-row">
        <span>Dark Mode</span>
        <div className="toggle on"></div>
      </div>

      <div className="setting-row">
        <span>Auto Alerts</span>
        <div className="toggle off"></div>
      </div>

      <button className="logout-btn">Log Out</button>
    </div>
  </div>
</div>
);
}

export default App;

```

13. References

- [1] World Health Organization. (2023). Epilepsy.
<https://www.who.int/news-room/fact-sheets/detail/epilepsy>
- [2] Fisher, R. S., Acevedo, C., Arzimanoglou, A., et al. (2014). ILAE official report: A practical clinical definition of epilepsy. *Epilepsia*, 55(4), 475–482.
- [3] Patel, S., Park, H., Bonato, P., Chan, L., & Rodgers, M. (2012). A review of wearable sensors and systems for monitoring health. *Journal of NeuroEngineering and Rehabilitation*, 9(21).
- [4] Teixeira, C. A., Direito, B., Feldwisch-Drentrup, H., et al. (2011). EPILAB: A software package for studies on the prediction of epileptic seizures. *Journal of Neuroscience Methods*, 200(2), 257–271.
- [5] Koren, J. P., Herta, J., Fürbass, F., et al. (2025). Automated seizure detection systems in clinical EEG monitoring: Performance and challenges. *Clinical Neurophysiology*.
- [6] Reus, M. A., et al. (2025). Clinical validation of EEG-based seizure detection software in epilepsy monitoring units. *Epilepsy & Behavior*.
- [7] Darankoum, J., et al. (2026). Deep learning approaches for seizure detection using EEG signals. *IEEE Transactions on Biomedical Engineering*.
- [8] Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.
- [9] Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.
- [10] Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach*
- [11] diagrams.net. (n.d.). *Online diagramming tool*. Retrieved from <https://app.diagrams.net/>
Roques, A. (n.d.). *PlantUML: Open-source tool for UML diagrams*. Retrieved from <https://plantuml.com/>